

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] DECEMBER 2024

Paper Code: CIC-209

Subject: Data Structures

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Differentiate between primitive and non-primitive data structures.	4	CO1
(b)	Define complexity. Explain Big-O notation with the help of a suitable example.	4	CO1
(c)	Calculate the address of element at $X[4, 3]$ in a 2D array $X[1..5][1..4]$ stored in row-major order. Assume base address is 1000 and each element requires 4 words of storage.	4	CO2
(d)	Write an algorithm/program to evaluate a postfix expression using a stack. Trace for: $(A + (B * C - (D / E * F) * G) * H)$	4	CO2
(e)	What is a threaded binary tree? Write the advantages of a threaded binary tree over a standard binary tree.	4	CO2
(f)	Write Kruskal's Algorithm for finding the minimum spanning tree of a weighted undirected graph.	4	CO3
UNIT – I			
2. (a)	What is meant by Row-Major Order and Column-Major Order in arrays? How do we calculate the address of an element in a row-major 2-D array?	5	CO2
(b)	Differentiate between a pointer to an array and an array of pointers with C syntax and memory diagrams.	5	CO1
3. (a)	How can a linked list be used to represent the polynomial $5x^3 + 4x^2 + 3x + 2$? Give an algorithm to perform addition of two polynomials using linked lists.	5	CO2
(b)	Write an algorithm/program to insert an element at the beginning of a circular linked list.	5	CO2
UNIT – II			
4. (a)	Define an AVL tree. What are the different AVL tree rotations? Construct an AVL tree for the following sequence of numbers: 10, 5, 8, 12, 18, 22, 1, 4, 6, 30	7	CO2
(b)	What are the fundamental properties and search invariants of a B-Tree?	3	CO2
5. (a)	Draw the unique Binary Tree when Preorder and Inorder traversals are given below, and write an algorithm for the same: Preorder: 1 2 4 8 5 3 6 7 Inorder: 4 8 2 5 1 6 3 7	5	CO2
(b)	Construct a Binary Search Tree (BST) by inserting the following sequence of numbers sequentially: 10, 12, 5, 4, 20, 8, 7, 15, 13	5	CO2

[P.T.O.]

Q. No.	Question	Max. Marks	CO(s)																									
UNIT – III																												
6. (a)	What do you mean by internal and external sorting? Sort the following elements using Bubble Sort showing each pass: 8, 22, 7, 9, 31, 19, 5, 13, 58, 93, 1, 45	5	CO2																									
(b)	Define Hashing. Why and how do collisions happen during hashing? Explain the different techniques used for resolving collisions in detail.	5	CO2																									
7. (a)	Write a complete algorithm/program for Binary Search on a sorted array.	5	CO2																									
(b)	Using the Binary Search algorithm, search for elements 23 and 47 in the given set of elements: [12, 23, 27, 35, 39, 42, 50]. Show all comparisons, mid calculations, and search boundaries at each step.	5	CO2																									
UNIT – IV																												
8. (a)	Draw the directed graph that corresponds to this adjacency matrix: <table><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>A</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>B</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>C</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>D</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>		A	B	C	D	A	1	0	1	0	B	1	0	0	0	C	0	0	0	1	D	1	0	1	0	5	CO3
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D	1	0	1	0																								
(b)	Give the complete algorithm for Breadth-First Search (BFS) graph traversal using a queue.	5	CO3																									
9. (a)	Describe the linked representation (adjacency list) of a graph with an illustrative example and node structures.	5	CO3																									
(b)	Define In-degree and Out-degree of a node in a directed graph using a suitable example.	5	CO3																									

*** End of Question Paper ***